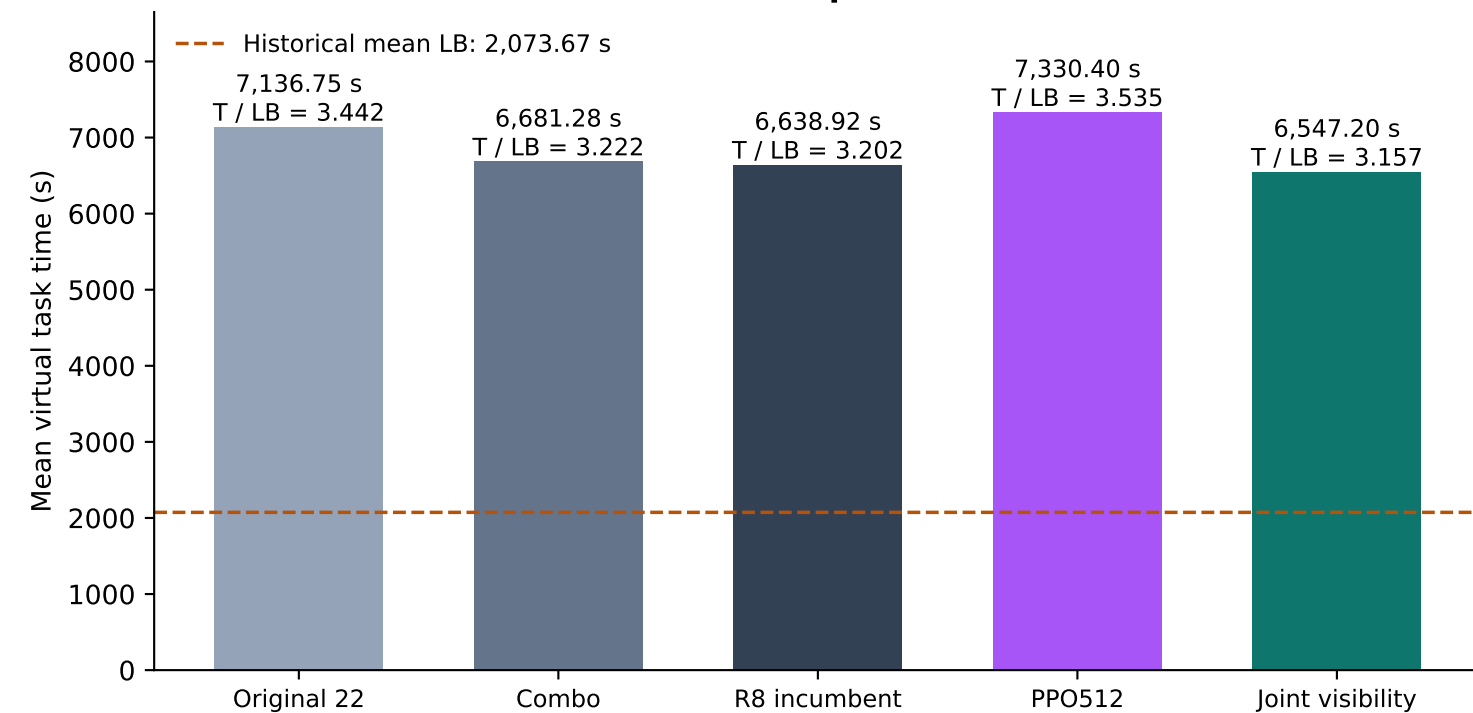
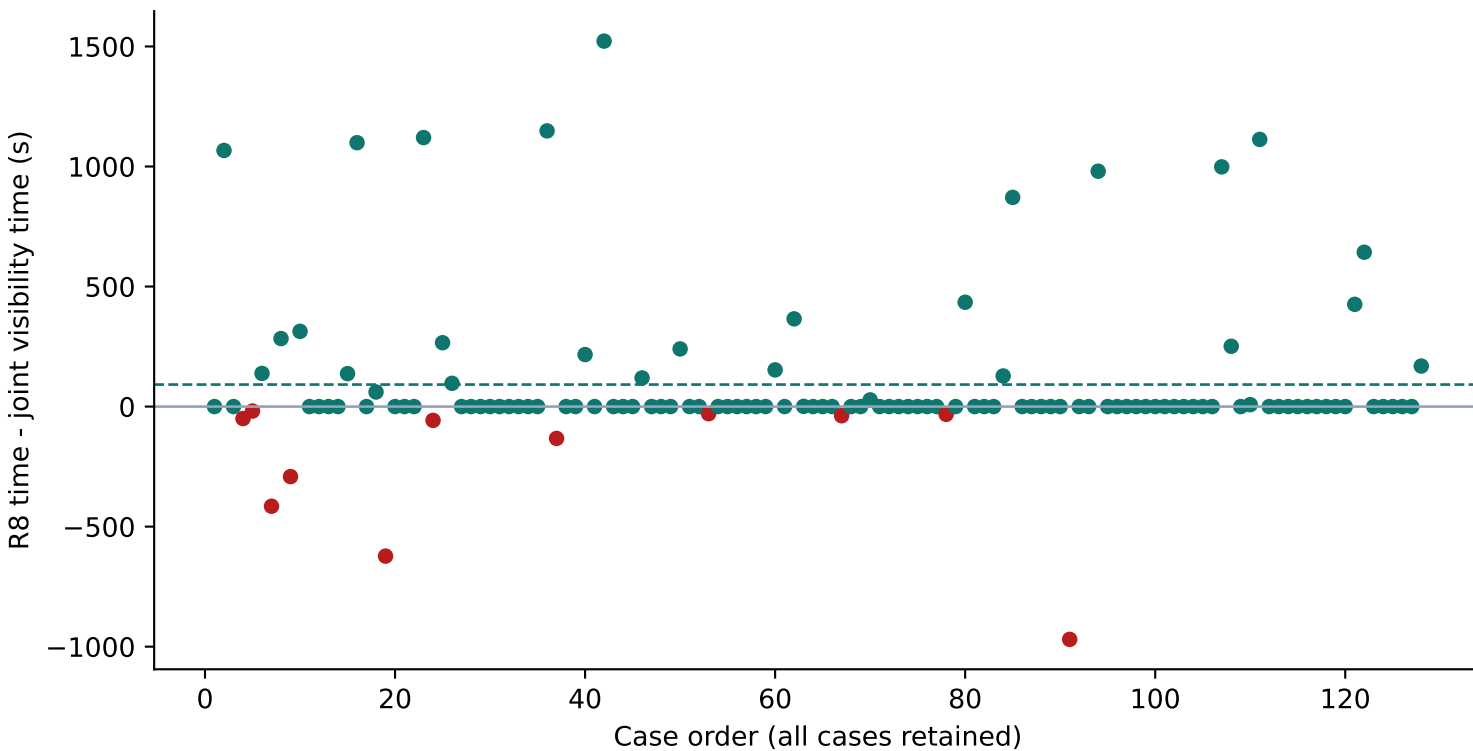


# Q4: joint positive and silent observations for localization

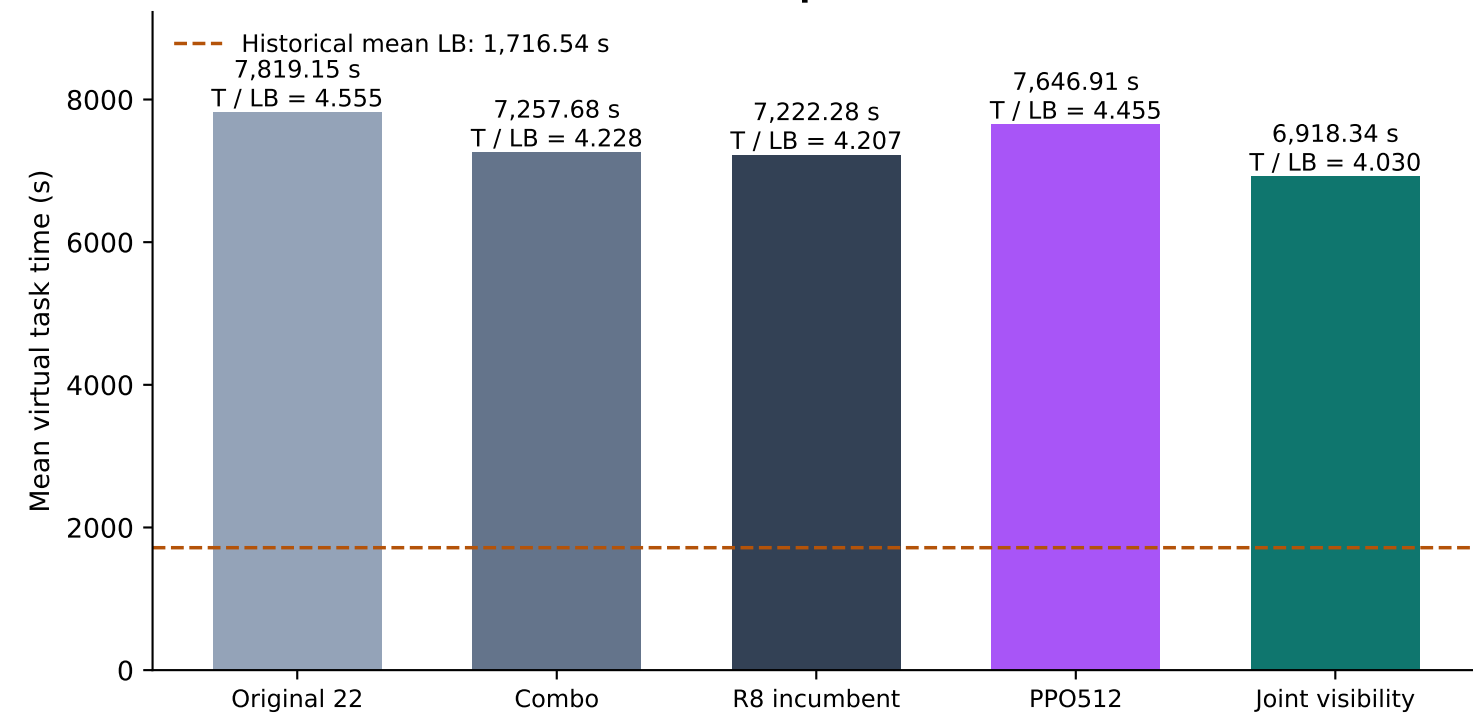
## Random: 128 paired cases



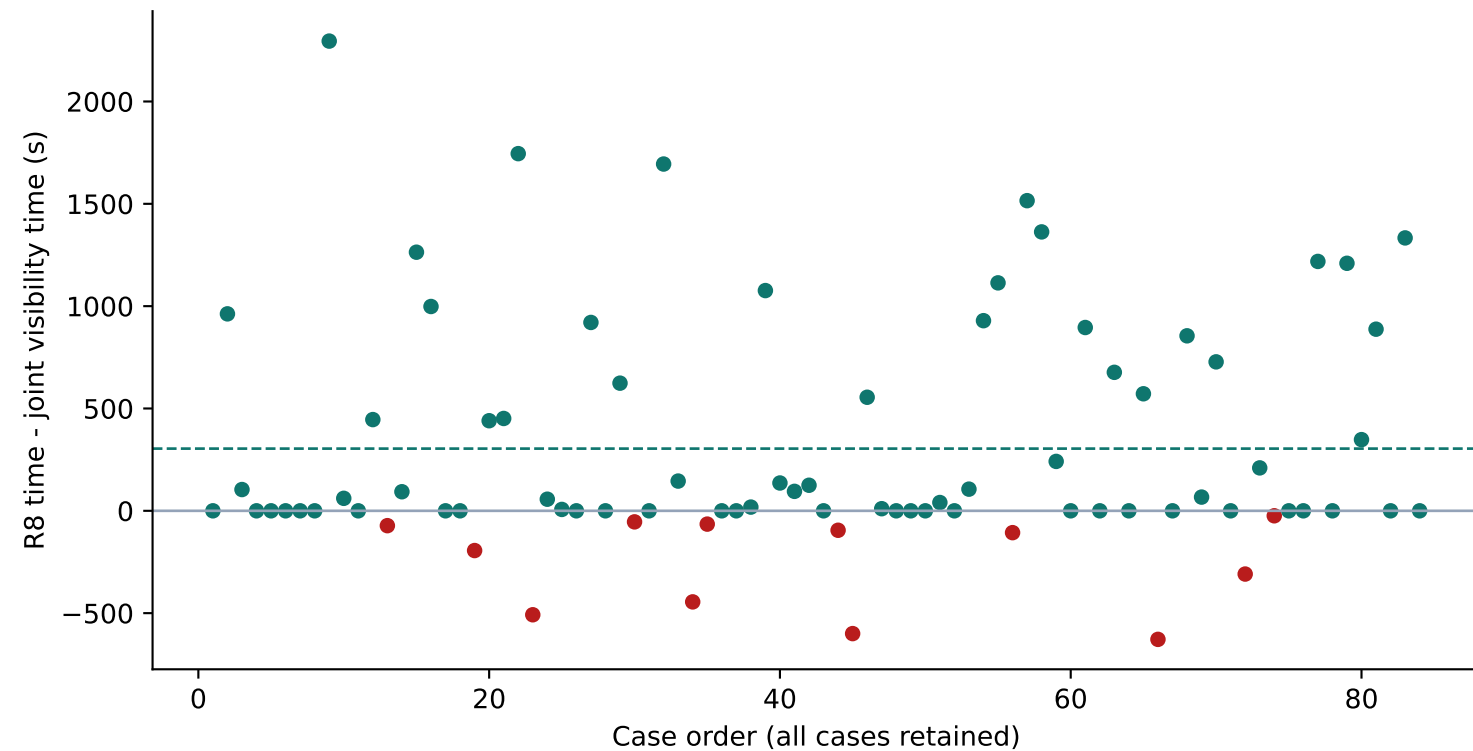
Mean saved 91.71 s (1.381%)  
Paired 95% CI: [39.22, 148.92] s



## Stress: 84 paired cases



Mean saved 303.93 s (4.208%)  
Paired 95% CI: [190.90, 429.61] s



Local generated mixed-source scenarios; all cases cleared. Historical oracle LB remains a relaxation, not an achievable-time claim.